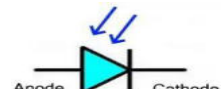
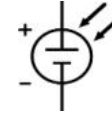
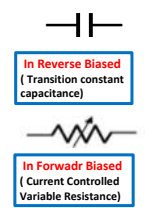
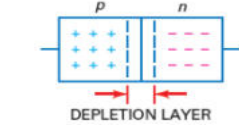
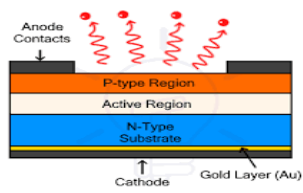
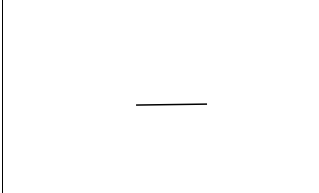
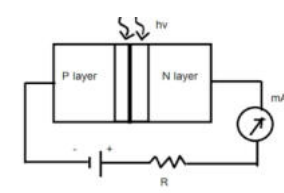
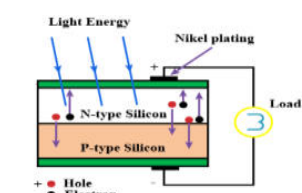
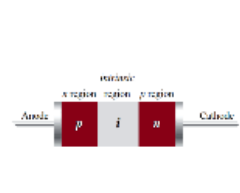
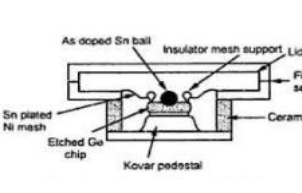
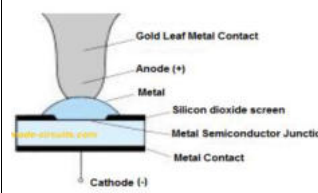
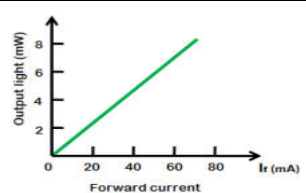
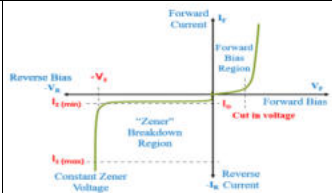
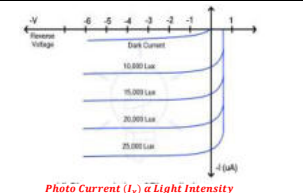
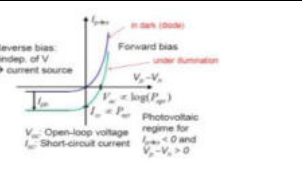
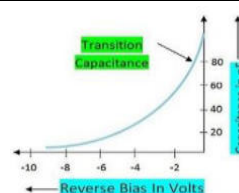
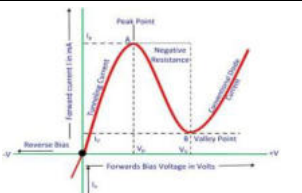
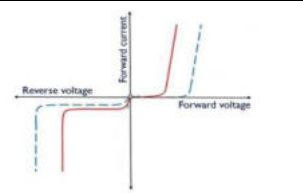
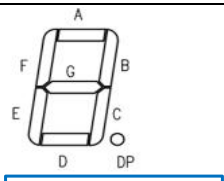
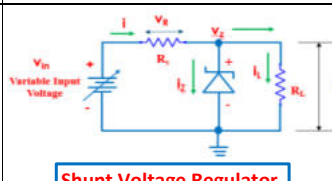
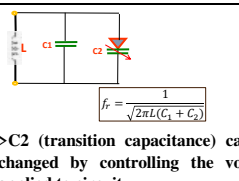


Special Types of Diodes & Transistors								
Types ➡	LED	Zener Diode	Photo Diode	Solar Cell	PIN Diode	Varactor Diode	Tunnel Diode	Schottkey Diode (Hot Carrier Diode)
Symbol								
Principle of Operation	> When LED is FB , it emit lights. > Recombination of electron and Hole pair gives away excess of Energy in the form of Light. (Electroluminescence)	> In the RB Condition , Used as Voltage Regulator . > Breakdown voltage 3 V to 200 V > Heavily Doping	> It is always operated in the RB conditions. > Dark Current :- The current flowing through a photo diode when there is no light incident on the device.	> Solar cell convert solar energy into electric energy > It works on the principle of " Photovoltaic Effect "	 In Reverse Biased (Transition constant capacitance) In Forward Biased (Current Controlled Variable Resistance)	 > Doped regions are like capacitor plate separated by a dielectric.	> Due to "Thin depletion region", an electron penetrates through the barrier. This is called tunneling and such high impurity density p-n junction device is called " Tunnel Diode "	> Special Diode having Very high Switching frequency (>10 Mhz) > Forward voltage drop (0.25 V) > Majority Charge Carrier
Construction								
Characteristics					> In Forward series resistance characteristics is graph of Rf Vs If > In Reverse capacitance characteristics is graph of Ct Vs Rv (Ct is Constant)			
Advantages	> Small Size & Light Weight > Long Life Compared to Bulb > High Brightness > High Operating Speed > Available in different spectral Colour	> Operated in RB > Heavily Doping > Breakdown voltage 3 v to 200 V	> High Speed of operation > High Sensitivity	> Do not require dc supply for operation > Longer life > Does Not Produce any Noise > Very fast response to incident of light > Not harmful for Environment	> Intrinsic region is sandwiched between n-type and p-type > Low forward resistance	> Variable capacitance in reverse voltage > Produce narrow depletion region	> Low cost > Low noise > High Speed > Low power consumption > Simplicity	> Very high efficiency > Low junction capacitance > Low forward voltage drop > Fast reverse recovery time so suitable for high speed switching > Operate at very high frequency (> 10 MHz) > Fast response to change in Bias > Low power consumption
Disadvantages	> Damage with over current > Luminous efficiency low > Require large power for operation	> Poor voltage regulation ratio > Cancel out voltage by applying larger voltage in reverse direction wasting electricity in the process	> Temperature Sensitive > Poor temp Stability > Dark current increase with temp > External bias voltage is required for operation	> Economical, Installation Cost is high > Conversion efficiency is low. > Temp Dependent > Operation depends on Daylight shining.			> Low output voltage swing > No isolation between I/P and O/P.	> Relatively high reverse current > Gets heated up quickly in practice. > Low reverse breakdown voltage > PIV rating is less compared to conventional diode.
Application	> In watches & Calculator > Seven Segment & Sixteen Segment Display > Infrared Remote Control > Indication Purpose in Different Application	> As Voltage regulator > In various protection circuit	> In Camera to Control light Intensity > In Fiber Optic Receiver > Object Counting System > In Alarm System	> Satellite & Space vehicles > Battery Charging System > Solar Cooker, Solar water heater > Automated street light, Emergency light. > Home lighting system	> Used as DC Controlled Microwave Switch. > Used as Photodetector in fiber optic systems.	> In LC tuned circuits > FM Modulators > Automatic frequency control devices > Television receivers > Parametric amplifier > Adjustable bandpass filter	> As a High speed switch > In pulse and digital circuits. > In Switching networks. > In negative resistance and high frequency oscillator. > In timing and computer logic circuits.	> Digital Computers > High Speed TTL (Transistor Transistor Logic) > Radar System > Detectors in communication system > Analog to Digital converter
Any One Application Figure	 Seven Segment Display	 Shunt Voltage Regulator				 $f_r = \frac{1}{\sqrt{2\pi L(C_1 + C_2)}}$ >C2 (transition capacitance) can be changed by controlling the voltage applied to circuit.		